

FiberNet

Comprehensive Fiber Network Research Toolkit

Project Status Report - 2026-07-05

GitHub: github.com/GellmanSparrowS/fibernet

Version: 1.22.0 (pyproject.toml) / 0.3.0 (setup.py)

Code: ~37,000 lines | Tests: ~10,500 lines | 889 test cases

1. Executive Summary

FiberNet is an open-source Python toolkit for fiber network structure research, targeting Nature Materials-level quality. It provides tools for generating, simulating, analyzing, and visualizing 2D/3D fiber network structures for eventual pip release.

Current Status: Developed to v1.22.0 with 68 generators, 20+ simulation modules, and 889 test cases. GitHub Actions CI/CD has failed ALL 47 consecutive runs due to a critical import error. Locally (with networkx), 887 pass and only 2 fail. Core functionality works but packaging issues need resolution.

Key Metrics

- * Source code: ~37,000 lines across 80+ Python modules
- * Test code: ~10,500 lines across 65+ test files
- * Generators: 68 (random, ordered, chiral, woven, bundles, curved, fractal, gradient)
- * Simulators: 20+ modules (FEM, dynamics, fracture, thermal, EM, acoustic, fluid, DMA)
- * Examples: 19 scripts covering major use cases
- * Local tests: 887 passed, 2 failed, 8 skipped
- * CI/CD: 47/47 runs FAILED (all Python versions x all platforms)
- * Commits: 30+ on main branch (2026-07-04)

2. Project Architecture

Clean modular architecture. Core defines data structures; specialized modules handle gen/sim/analysis/viz.

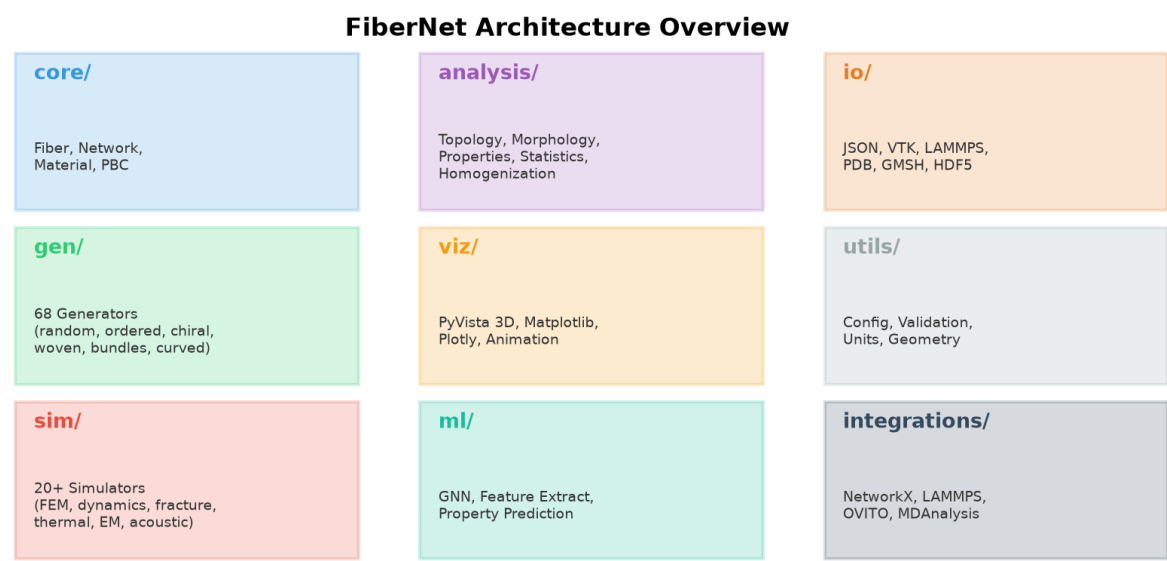


Figure 1: FiberNet modular architecture

Module Breakdown

- * core/ (7): Fiber, FiberNetwork, Material, PBC, Transform, Copy, Crosslinks
- * gen/ (12): ordered, disordered, chiral, woven, hierarchical, bundles, curved, fractal, gradient, specialized, variants, advanced
- * sim/ (24+): mechanical, dynamics, fracture, damage, thermal, EM, acoustic, fluid, viscoelastic, creep, fatigue, diffusion, DMA, rheology, buckling, multiscale, nonlinear, coupled, periodic, accelerated, cte, permeability, optimization, uncertainty
- * analysis/ (11): topology, morphology, properties, statistics, advanced, stress_strain, percolation, spatial, homogenization, comparison, networkx
- * viz/ (8): plot2d, render3d, animate, plotly, advanced, visualization, pyvista, trimesh
- * ml/ (4): dataset, features, predictor, graph_neural_network
- * io/ (8): vtk, pdb, xyz, lammps, gmsh, pandas, fea_export, mesh_export
- * utils/ (9): config, validation, units, geometry, io, batch, ensemble, parametric, exceptions
- * integrations/ (4): networkx, lammps, ovito, mdanalysis

3. Generated Network Structures

68 generators covering diverse fiber network architectures:

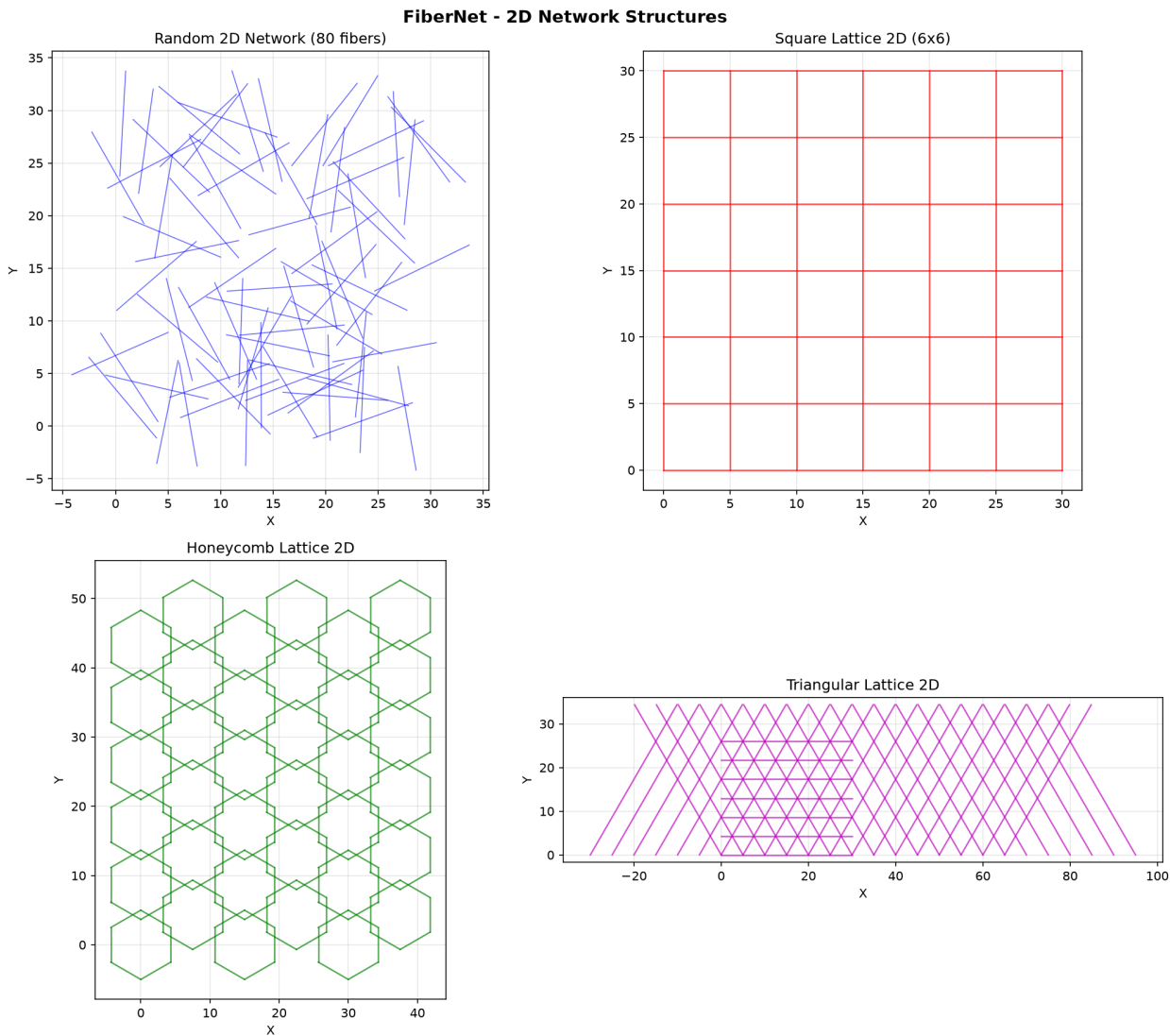


Figure 2: 2D - random, square, honeycomb, triangular

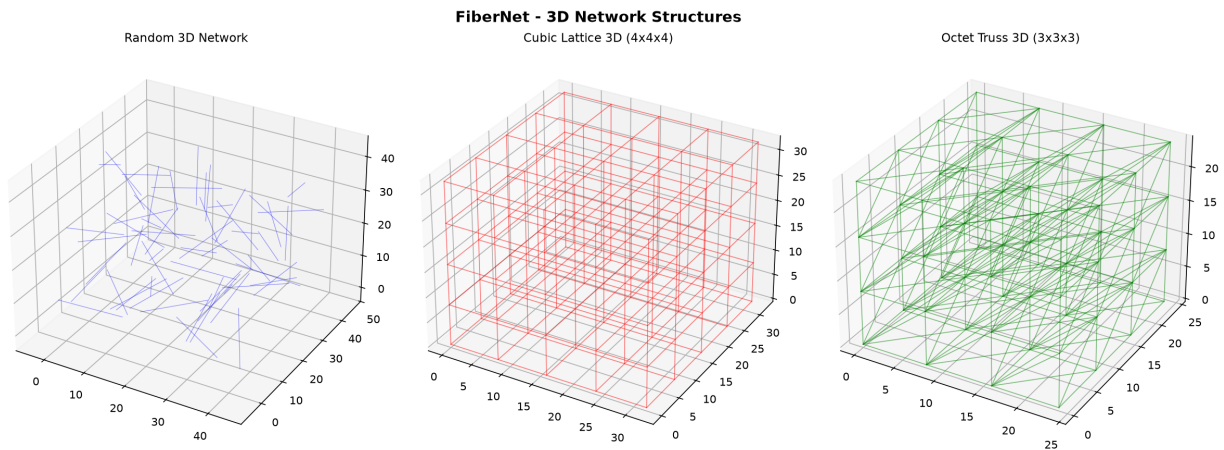


Figure 3: 3D - random, cubic lattice, octet truss

Generator Categories

- * Random (8): straight/walk 2d/3d, curved, electrospun, paper, CNT
- * Ordered (9): square, honeycomb, triangular, kagome (2D); cubic, BCC, FCC, octet, diamond (3D)
- * Chiral (1): chiral_network_2d with tunable twist
- * Woven (5): plain, twill, satin, knitted, nonwoven
- * Hierarchical (1): multi-level structure
- * Bundles (7): parallel, twisted, random, braided, tendon_like (2D/3D)
- * Curved (9): sinusoidal, helical, arc, Bezier; random_curved, crimped
- * Specialized (28): fractal, gradient, biomimetic, domain-specific

4. Simulation Capabilities

18+ physics domains, 20+ dedicated modules:

- * Mechanical/FEM: Beam FEM, stretching+bending, stress-strain, incremental FEM
- * Dynamics: MD (Verlet, Langevin), time-dependent loading
- * Fracture: Damage mechanics, Griffith criterion, SIF
- * Thermal: Steady/transient conduction, conductivity
- * Electromagnetic: Permittivity, conductivity, effective medium
- * Acoustic: Wave propagation, band structure, phononic
- * Fluid: Darcy flow, pore network, permeability
- * Viscoelastic: Maxwell/Kelvin-Voigt models
- * Creep/Fatigue: Long-term deformation, S-N curves
- * DMA: Storage/loss modulus
- * Buckling: Eigenvalue analysis, stability
- * Diffusion: Mass transport, Fickian
- * Multiscale: Hierarchical modeling
- * Optimization: L-BFGS-B, CG, BFGS, Powell, DE
- * Uncertainty: Monte Carlo UQ
- * CTE: Thermal expansion | Rheology: characterization

5. Test Results

5.1 Local Tests (with networkx)

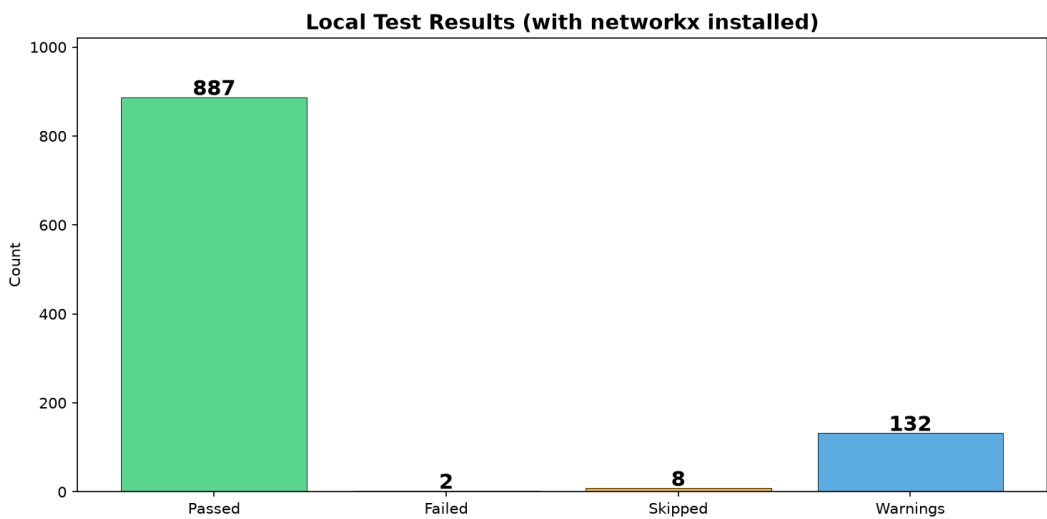


Figure 4: Local test results

887 PASSED | 2 FAILED (fea_export) | 8 SKIPPED | 132 warnings | ~2.5 min

5.2 Failed Tests

Failure 1: test_export_node_count
AssertionError: assert 113 == 60
Curved fiber discretization produces more nodes.

Failure 2: test_export_empty_network
IndexError: too many indices for 1-dim array
Missing empty guard at fea_export.py:374

6. GitHub Actions CI/CD

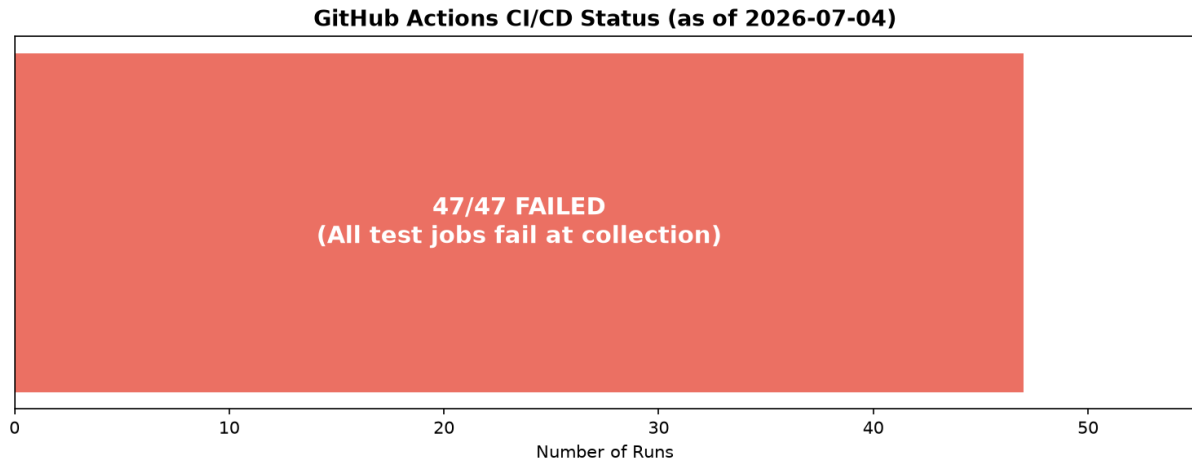


Figure 5: 47/47 runs failed

6.1 Root Cause

All test jobs fail at pytest collection. Import chain:

1. fibernet/__init__.py imports fibernet.analysis
2. analysis/__init__.py unconditionally imports topology
3. topology.py: import networkx fails, nx undefined
4. Line 113: "-> nx.Graph:" eager annotation crashes
5. NameError propagates, all 46 test files fail

6.2 Why networkx fails

CI: pip install -e ".[dev,full]"

pyproject.toml: networkx in "full" extras (correct)

setup.py: NO networkx (v0.3.0 vs pyproject v1.22.0)

Dual config causes pip to use wrong metadata

6.3 Fixes (Priority)

1. (CRITICAL) "from __future__ import annotations" in topology.py
2. (CRITICAL) Remove setup.py, keep pyproject.toml
3. (HIGH) try/except in analysis/__init__.py
4. (MEDIUM) Fix fea_export.py empty guard
5. (LOW) Fix test_export_node_count

7. Dependencies & Licensing

Core (all permissive)

- * numpy $\geq$ 1.21 (BSD)
- * scipy $\geq$ 1.7 (BSD)
- * networkx $\geq$ 3.0 (BSD)
- * matplotlib $\geq$ 3.5 (PSF/BSD)

Optional

- * pyvista (MIT) | h5py (BSD) | taichi (Apache 2.0)
- * scikit-learn (BSD) | trimesh (MIT) | plotly (MIT)
- * LAMMPS (GPL v2) - must remain optional
- * OVITO (GPL v3) - must remain optional

MIT License compatible with all deps. ATTRIBUTIONS.md credits third-party code.

8. Gap Analysis

Generation

- * [DONE] 2D/3D random, ordered, chiral, woven, bundles, curved
- * [PARTIAL] Compression/kink bands
- * [MISSING] Tubular braids generator

Simulation

- * [DONE] FEM, dynamics, fracture, thermal, EM, acoustic, viscoelastic
- * [PARTIAL] Multi-physics coupling (not validated)
- * [PARTIAL] GPU acceleration (Taichi, has warnings)
- * [BUGGY] Abaqus export (2 failures)
- * [MISSING] Published benchmarks

Packaging

- * [BROKEN] CI/CD: 47/47 failed
- * [BROKEN] setup.py vs pyproject.toml
- * [MISSING] PyPI, DOI
- * [PARTIAL] Documentation (Sphinx OK)
- * [DONE] MIT License, CONTRIBUTING.md

9. Recommendations

Immediate

1. Fix topology.py annotations
2. Remove setup.py
3. Guard imports
4. Fix fea_export
5. Verify CI

Short-term

6. Integration tests
7. Validate vs analytical
8. Annotations audit
9. Zenodo DOI
10. README gallery

Medium-term

11. More generators
12. Multi-physics validation
13. GPU benchmarks
14. Benchmark paper
15. Jupyter tutorials
16. ReadTheDocs

Long-term

17. Reproduce literature
18. Bio fiber models
19. Multi-scale
20. Community
21. Software paper
22. ML prediction

Appendix: Recent Commits

4d0ff7a	22:42	Add progress summary document
24dc1e5	22:37	v1.22.0: SciPy optimization
acfe89c	22:33	v1.21.0: Curved fiber generators
31e7d64	22:30	v1.20.0: Dynamics + validation
65bcf27	22:25	v1.19.0: Bundles, PyVista, trimesh
db8c7f9	22:04	Example 15: ML prediction
219b1ed	22:01	Comprehensive CHANGELOG.md
940e9ad	21:55	v1.18.0: Mesh export
3cf44ed	21:49	v1.17.0: Network comparison
0c633b9	21:46	v1.16.0: Homogenization
a50bce7	21:42	v1.15.0: Spatial statistics
8334a06	21:31	v1.14.0: Material database
b84a077	21:22	v1.13.0: Fractal & gradient
dba2ea2	21:08	v1.12.0: Topology analysis
728c5a0	20:54	v1.11.0: Viz & DOE modules